

Writing the SRS.

From skeleton to a complete Software Requirements Specification — 15 functional, 5 non-functional, full traceability, ready for M3 grading.

LECTURE · 90 MIN

M3 Mandatory milestone

Followed by SRS lab — 90 min practice on your own Leitprojekt.

Today's Agenda

90-min lecture + 90-min lab — first Mandatory milestone

00–10

Entry quiz & M2 template check

Two-minute quiz + round-up of last week's template drafts.

10–35

SRS skeleton in detail

Scope, functional & non-functional requirements, external interfaces.

35–60

Formulation training by Rinzler

User stories → testable requirements · must / should / could priority.

60–85

AI workshop preparation

Prompt templates for requirement extraction, the Use → Verify → Cite protocol.

85–90

M3 deliverables and grading

15 functional + 5 non-functional requirements, traceability rows.

Recap: Week 02

Three questions on standards — three minutes, written, then shared.

QUESTIONS

- 1. Which ISO 29148 attribute did we name most often?**
- 2. Why is “user-friendly” not yet a requirement?**
- 3. Give one example from M2 where you had to choose SRS vs. SDD.**

WHY RETRIEVAL OPENS EVERY LECTURE

Retrieval practice roughly doubles long-term retention at twelve weeks (Roediger & Karpicke, 2006). Three minutes spent here is cheaper than half an hour of re-teaching in W8.

In this course, retrieval doubles as a gap check. If more than half the room misses question 1, we extend ISO 29148 coverage next time.

SRS Skeleton in Detail

Every section exists to answer a specific question — skip a section, lose the answer.

SECTIONS AND THE QUESTION EACH ONE ANSWERS

- 1. Introduction (1.1 Purpose · 1.2 Scope · 1.3 Audience)** — answers: for whom, why now, what is in vs. out of scope?
- 2. Overall description** — answers: what context does the system live in (users, hardware, dependencies, constraints)?
- 3. Functional requirements** — answers: what the system shall do, one numbered requirement per behaviour, with input / action / observable output.
- 4. Non-functional requirements** — answers: how fast, how reliable, how secure, how maintainable — with measurable boundaries.
- 5. External interface requirements** — answers: which protocols, data formats, hardware ports?
- 6. Other requirements** — answers: legal, regulatory, audit, localisation.
- 7. Appendices · glossary, traceability matrix, assumptions** — answers: what terms mean and how each requirement is verified.

From Story to Testable Requirement

User story → precise verb → input / action / observable output → priority.

STORY (RINZLER, WEAK)

“As a researcher, I want to upload a PDF so that I can search inside it.”

TESTABLE REQUIREMENT (RINZLER, REPAIRED)

FR-12 (should) · The system shall accept a PDF of up to 20 MB and report in JSON {ok, pages, indexedAt} within 5 s, or {error, code} within 2 s, on the standard endpoint POST /docs. Failure boundary: 20 MB is rejected with HTTP 413.

PRIORITY — MUST / SHOULD / COULD

must — system fails without it (MoSCoW “Must”). Test failure blocks release.

should — stakeholder pain if missing, business case for it. Ship with planned close-out.

could — nice-to-have, post-release roadmap.

TRACEABILITY ROW — EACH REQUIREMENT

ID FR-12 · **Source** interview w/ Research group 2026-03. · **Verifies** integration test IT-PDF-01 · **Owner** T. Baasan.

Without traceability, an SRS is a wish list. With it, every line maps to a test, an owner, and a source interview.

CONCEPT TEST

Vote individually · 30-second peer defence · re-vote · reveal

QUESTION 1 / 1

You have just received a 2-page stakeholder brief. Which of these requirements is closest to ISO 29148's *Verifiable* attribute?

A

"The system shall be fast."

B

"The system shall respond quickly."

C

"The system shall answer 95% of GET /search calls within 200 ms."

D

"The system shall be performant."

REVEAL ANSWER — AFTER VOTES

C is correct.

A black-box load test can measure 95% / 200 ms. None of the others name input, action, output or boundary. This is the only one you could write a passing/failing test for today, even before any code exists.

M3 — the First Deliverable That Counts

SRS v1.0 in your workspace by Friday — the rest of the semester builds on it.

MINIMUM CONTENT — WHAT GRADERS CHECK

15 functional requirements — each with ID, source, priority (must/should/could), verification method, owner.

5 non-functional requirements — on performance, security, reliability, maintainability, accessibility.

Traceability matrix — one row per requirement, linked to source interview and to a planned test.

Human-vs-AI comparison table — one column for your draft, one for the AI output from the upcoming lab.

Published in your workspace — URL for grading; PDF export optional.

GRADING RUBRIC — M3

- 30% Completeness (count vs. declared)
- 25% Testability (peer rewrites each requirement as a black-box test)
- 20% Quality attributes (Unambiguous, Complete, Verifiable — explicit checklist)
- 15% Traceability rows
- 10% Human-vs-AI table + reflection

Pass rule: M3 must be submitted; otherwise no overall grade — even in the retake case.

Take-aways & Next Week

The SRS is the contract you write with your future self.

THREE TAKE-AWAYS FROM WEEK 03

1. Every requirement is id + source + priority + verification method + owner.
2. Non-functional requirements are mandatory — five is just the floor.
3. AI generates text; only the verification protocol turns it into a specification.

M3 DELIVERABLE · DUE FRIDAY

SRS v1.0 in your workspace. 15 functional + 5 non-functional. Traceability rows. Mensch-vs-KI table.

NEXT WEEK · WEEK 04

Architecture documentation.

Once the system's contract is on paper, we describe its shape. arc42 gives the prose template, C4 the diagram levels. ADR capture makes decisions legible for the next joiner — a direct counter-measure to tribal knowledge (W1 anti-pattern 01).